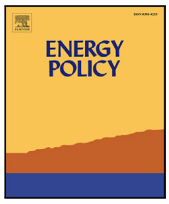
Energy Policy 201 (2025) 114556



Contents lists available at ScienceDirect

Energy Policy

journal homepage: www.elsevier.com/locate/enpol





International trade barriers, export and industrial resilience: An empirical study based on the EU and USA antidumping and countervailing policies on photovoltaic products☆

Mudan Wang, Zeyu Xie *

Research Institute for Eco-civilization, CASS, No. 27 Wangfujing Street, Dongcheng, 100710, Beijing, China

ARTICLE INFO

Keywords:
Trade barriers
Antidumping and countervailing policies (ACPs)
Export resilience
Firm entry
Technological innovation

ABSTRACT

The increasing trade barriers imposed by the EU and the USA on China have created a challenging environment for the growth of Chinese emerging and high-tech industries. This paper, which is based on new trade theory with substitutes, examines how antidumping and countervailing policies (ACPs) from the EU and the USA impact Chinese photovoltaic exports and firm numbers. Using product-level trade data, business registration data, and patent application data, a staggered DID model is employed to assess the effects on exports, firm entry, and innovation. The results show that ACPs resulted in a significant decrease in short-term Chinese PV exports of approximately 87.5% and a reduction in the number of new entrants of approximately 10.7%. However, in the long term, both exports and firm entry exhibited a "V-shaped" rebound, with firm entry rebounding faster and more significantly. Further analysis revealed that ACPs led to a rapid increase in domestic photovoltaic capacity and a sustained rise in patent applications, highlighting the effectiveness of expanding the domestic market and focusing on technological innovation to counter international trade barriers.

1. Introduction

Climate change mitigation is a common challenge faced by mankind, and in the context of economic globalization and the internationalization of the industrial division of labor, countries should fully consider their respective advantages and make concerted efforts to develop green and low-carbon industries and promote carbon emission reductions. However, in recent years, trade friction between major economies has been increasing (Metiu, 2021), especially Europe and the United States, which have repeatedly introduced trade barriers against the new Chinese energy industry, such as antidumping and countervailing measures for the photovoltaic (PV) industry, the U.S.-China trade war, the EU Carbon Border Adjustment Mechanism (CBAM) and trade probe into new energy vehicle. Additionally, the foreign trade blockade of China will not only seriously impact Chinese exports, which will have a negative impact on the healthy development of Chinese-related industries and optimization and upgrading but also increase the cost of the global development of new energy, hindering the process of energy decarbonization. Such an approach is not conducive to Chinese industrial security and trade security. In this context, studying the short-term and long-term impacts of trade barriers on Chinese exports and the development of related industries, exploring whether trade barriers in Europe and the United States impede the development of the Chinese new energy industry and discussing its impact on climate change are highly practical and valuable tasks.

Currently, China is the world's leading producer and exporter of photovoltaic products, having established a comprehensive PV industry that includes silicon mining and refining, ingot/wafer production and cutting, cell manufacturing, module encapsulation, power generation, auxiliary material production, and supporting facilities, holding a significant position in the global photovoltaic market. However, more than a decade ago, the development of the Chinese PV industry was severely restricted by the reliance on foreign raw materials, markets and equipment. Europe and the USA, as the main export markets for Chinese PV products, successively introduced antidumping, countervailing and other trade-restrictive policies for Chinese PV products from 2011–2012, leading to a significant increase in Chinese PV trade costs and a

☆ This research was funded by the National Social Science Fund of China (Grant No. 23AZD060) and Research Institute for Eco-civilization, CASS-Guizhou Academy of Social Sciences Collaborative Project "Research on Green and Low Carbon Development Strategies and Policies in Guizhou Province" (Grant No. STWM-HX-2023-012).

* Corresponding author.
E-mail addresses: wangmd@cass.org.cn (M. Wang), xiezy@cass.org.cn (Z. Xie).

https://doi.org/10.1016/j.enpol.2025.114556
Received 17 October 2024; Received in revised form 2 February 2025; Accepted 18 February 2025
Available online 7 March 2025
0301-4215/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

serious reduction in exports. The profit margins of domestic firms were squeezed, and representative firms such as Wuxi Suntech even went bankrupt and restructured due to enormous losses. At that time, ACPs of the EU and the USA cast a thick shadow over the Chinese PV industry. However, to everyone's surprise, by 2022, Chinese PV technology and manufacturing capacity had reached a world-leading level, and Chinese patents far exceeded those of other countries, accounting for 80% of the global total, with obvious technical advantages and a high level of competitiveness. Moreover, more than 70% of the world's PV products come from China. By 2022, PV modules and other products from China had been top ranking globally for 16 years, with both the export value and volume continuously increasing. On this basis, the trade protection actions of Europe and the USA seem to have promoted the development of the Chinese PV industry, which is astonishing. This paper focuses on the rebirth of Chinese PV industry from the ACPs of the EU and the USA to acquire policy insights for China to respond reasonably to trade barriers and other external risks, enhance trade resilience, and promote industrial optimization and upgrading.

However, there are still difficulties associated with conducting research on the aforementioned topics. First, it is not easy to obtain precise product-level trade data, making it difficult to directly analyze how trade barriers affect exports related to PV products. Second, owing to the limited availability of microlevel data, there is a lack of detailed information on firms and patents in the PV industry, which hinders in-depth analyses of corporate behavior adjustments given the impact of trade barriers. To address these issues, product-level data on Chinese PV trade and data on PV-related antidumping and countervailing cases for 2009–2019 are obtained from the Trade Map database developed by the World Trade Organization (WTO), the United Nations Conference on Trade and Development (UNCTAD), the International Trade Centre (ITC), and the World Bank's Temporary Trade Barriers database. The difference-in-differences (DID) model and the event study method are applied to assess the short-term and long-term policy effects of ACPs of the EU and the USA on Chinese PV trade and industry development in approximately 2011. The results reveal that the ACPs of the EU and the USA led to a significant decline in Chinese PV exports, and this adverse impact exhibited noticeable temporal heterogeneity, resulting in a sharp reduction in Chinese PV exports to Europe and the USA in the short term, with the negative impact gradually diminishing in the long term. Furthermore, over 39.4 million corporate registration data points for 2009–2018 are obtained from the Qichacha website and 19.3 million patent application records are obtained from the China National Intellectual Property Administration (CNIPA) database to systematically evaluate the impact of ACPs on the market vitality and technological innovation of the Chinese PV industry at the county–industry level. The results indicate that the ACPs of the EU and the USA had a significant negative impact on new market entities in the Chinese PV industry in the short term. However, only two years after the introduction of the trade barrier in 2014, the Chinese PV industry had already recovered from the downturn, with market vitality displaying rapid recovery. An analysis based on PV patent applications reveals that ACPs never had a significant effect on patent applications in the Chinese PV industry, implying that adherence to innovation, research and development was an important reason for the effective response to external challenges.

The first contribution of this paper is an analysis of the impact of trade barriers on exports from sanctioned industries and market entry under the condition of existing substitutes, as well as the effective ways for enterprises to cope with trade barriers. It serves as a beneficial extension to heterogeneous firm trade theory and trade policy-related research. For a long time, the export suppression effect and welfare effect of trade barriers have been popular topics in academia. The serious inhibitory effect of trade barriers such as antidumping and countervailing duties on exports has also been confirmed by empirical studies (Jabbour et al., 2019; Cheng et al., 2021). However, when a industry in a sanctioned country can maintain a technological advantage, its marginal production cost will continue to decline, partially or fully offsetting the cost premium associated with trade barriers. In such a scenario, it is necessary to explore whether trade barriers have a temporally heterogeneous impact on relevant industry exports in the sanctioned country. In addition, the literature mainly discusses the effects of trade barriers on national trade or on the exports of all enterprises. Industry-level studies focus mainly on the broad manufacturing industry (Grundke and Moser, 2019; Nguyen and Kinnucan, 2019), and discussion of the exports from more segmented industries is limited. In ACPs studies, input–output models have been applied to explore the effects of trade barriers on Chinese participation in the global value chain (Wang et al., 2019), regression models have been used to explore within-firm input reallocation resulting from trade protection (Vandenbussche and Viegelahn, 2018), and analyses of the transmission and diffusion effects of Sino-US trade friction along global value chains have been performed (Wang and Shi, 2022); however, the heterogeneous temporal effects of antidumping and countervailing duties on Chinese PV exports have received little attention. To fill the gaps in the literature, the substitution relationship between PV products and fossil fuel products is embedded into a theoretical trade model, and the effects of the EU and the USA ACPs on Chinese PV exports, market entry thresholds, and the number of enterprises are explored, providing a theoretical basis for responding to trade barriers by expanding the domestic market and strengthening technological innovation. Then, at the empirical level, on the basis of HS 8-digit product-level trade data, a DID model is applied to explore the short-term trend of Chinese PV exports to Europe and the USA after the ACPs, and an event study method is adopted to examine the corresponding long-term adjustment direction, revealing the heterogeneous impact of ACPs at different time scales and the "V-shaped" reverse form of Chinese PV exports after trade barriers. This paper provides a novel analysis of the effects of the EU and the USA ACPs from a microperspective.

This paper makes a second contribution to the literature by exploring the short-term and long-term impacts of trade barriers on the Chinese PV industry, specifically regarding how these barriers inhibit global specialization and the division of labor. By utilizing fine-scale Chinese business enterprise registration data, our analysis extends the findings in the literature, which predominantly focuses on the effects of trade barriers on exports (Blonigen and Prusa, 2016; Felbermayr and Sandkamp, 2020). While some studies have further explored the impact of trade barriers on exporters' productivity, aggregate output (Vandenbussche and Viegelahn, 2018), firm profits (Fontagné et al., 2015), stock market performance (Huang et al., 2023), resource allocation (McCarthy, 2016), and employment (Li and Whalley, 2021), there remains a gap in the literature regarding the impact of such barriers on sanctioned countries, particularly from the standpoint of how they inhibit specialization within the division of labor and the entry of firms into relevant industries. The most closely related studies (Lu et al., 2013; Crowley et al., 2018) have examined the effects of trade policies on Chinese firms' entry into and exit from export markets, but these studies did not address the implications of trade policies for firms' entry into and exit from domestic product markets. However, trade barriers may lead firms not only to exit export markets but also to cease operations entirely. To address this gap, our study leverages a dataset of more than 39.4 million Chinese industrial and commercial enterprise registrations from 2009–2018. Through text analysis of the business scope within these registration records, we identify firms within the PV industry chain, thereby overcoming the limitations of broadly categorized industry information in standard registration data. Our findings reveal that the European and American antidumping and countervailing measures significantly reduced the number of new market entrants in the Chinese PV industry in the short term. Despite these trade barriers, China, the world's largest market for newly installed renewable energy capacity, has provided substantial domestic demand that has enabled rapid "V-shaped" recovery in the number of new local

PV firms, demonstrating industry resilience. Given that the new energy sector is a highly emergent industry reliant on R&D investment and that trade barriers increasingly target R&D-intensive industries (Kao and Peng, 2016), this paper further investigates the impact of European and American trade barriers on technological innovation within the Chinese PV industry. Based on an analysis of 19.3 million patent applications, we find that the number of patent filings in the Chinese PV sector has continued to grow despite technological embargoes. This study not only contributes to the literature on the impact of trade barriers on industrial technological innovation but also underscores the critical roles of the domestic market and technological advancement in mitigating the effects of international trade barriers.

The remainder of this paper is organized as follows: in the Section 2, the policy background and theoretical analysis are presented; the Section 3 details the research design; in the Section 4, the empirical results and analysis are presented; the Section 5 provides further discussion; and finally, conclusions and policy recommendations are presented in the Section 6.

2. Policy background and theoretical analysis

2.1. Policy background

Promoting the application of clean technology, such as solar power generation, and accelerating the green transformation of energy structure are inevitable choices for countries worldwide to address the dual challenges of energy crisis and climate change. PV product trade is an important step in the acquisition of high-quality renewable energy products and the promotion of renewable energy technology globally, especially for economies with growing consumption demand or insufficient domestic product supply, and the development of solar power generation is highly important. Over the past decade, PV product trade has displayed a fluctuating growth trend. According to the statistical data,[1] the export value of global PV cells and modules increased from approximately 54.7 billion USD to approximately 74.1 billion USD from 2012–2021, a growth of 35.5%, which was greater than that of global merchandise trade during the same period (20.3%). Currently, China produces PV modules that account for approximately 70% of the total global market, and the center of the world's PV industry is gradually shifting toward China.

However, the rapid development of the Chinese PV industry has not been smooth. As China has secured a place in the international PV market, increasingly frequent trade frictions have continuously disturbed the Chinese PV industry. Between 2011 and 2021, a total of 16 trade remedy investigations involving Chinese PV companies were launched.[2] The China-centric ACPs issued by developed economies such as those in Europe and the USA are the most typical, characterized by long-lasting cases, numerous reviews, and high punitive tariffs. In recent years, investigations and sanctions against Chinese PV products have also been initiated in developing economies such as India and Turkey. The Chinese PV industry faces trade risks and international environmental uncertainties that cannot be ignored.

The USA took the lead in launching ACPs against Chinese PV exports in 2011 and imposed a high duty on Chinese products in 2012. In 2014, the USA launched another ACPs against China and made a positive final ruling in one investigation in December of the same year. During the implementation of trade investigations and related measures, administrative reviews and reviews of changes in circumstances occurred, and most of the Chinese products involved sustained substantial limitations, with the extension of USA-related measures multiple times. From 2017–2019, the USA initiated Section 201 Action, Section 301 Action, and Section 337 Action, all of which were related to PV products. In 2022, the USA announced a four-year extension of the 201 safeguard duty and granted Mexico and Canada a duty-free route. In 2023, the USA announced that it would launch Section 337 investigations against the producers of Chinese PV connectors and related components. EU-China PV trade sanctions can be traced back to the ACP in 2012. Although Europe and China reached a minimum import price limit agreement in 2013, the EU still imposed antidumping and countervailing duties on Chinese PV companies that were not party to this price agreement. In September 2018, the EU formally terminated ACPs against Chinese PV products.

The investigation processes of antidumping and countervailing duties in Europe and the USA are relatively similar and involve stages such as the initiation of investigation, preliminary determination, final determination, and termination of measures. Here, with the United States as an example, the antidumping and countervailing investigation procedure is described. In the USA, the Section 701 Tariff Act of 1930 provided the fundamental legal basis for antidumping and countervailing investigations. The United States International Trade Commission (USITC) and the Department of Commerce(DOC) are the two national agencies responsible for conducting antidumping and countervailing investigations, as well as sunset reviews after five years. Domestic industries can apply to the aforementioned agencies for relief investigations against imported products that are sold at less than fair value (i.e., dumping) or receive actionable funding provided by foreign governments (i.e., subsidies) on the basis of this law. The aforementioned agencies should decide whether to file a case within a specified time (20 days after receiving an application). During the investigation process, the USITC and the DOC have different responsibilities. The former determines whether the domestic industry has been materially injured by the imported products under investigation, whereas the latter determines whether dumping or subsidizing behavior actually exists. If the result of an investigation is affirmative, the corresponding dumping margin or subsidy amount must be determined. The USITC usually needs to make a preliminary determination within 45 days after receiving an investigation application, and the DOC issues a preliminary determination within 40 days (antidumping cases) or 115 days (countervailing cases) after the preliminary determination of the former. If a positive preliminary determination is made, the defendant importer should pay a cash deposit or bond according to the dumping margin or subsidy amount; otherwise, the case investigation is terminated. Distinguishing between antidumping and countervailing cases, the USITC and the DOC generally make a final determination on a case within 280 days after receiving the trade remedy investigation application. In cases with positive determinations, antidumping duties, countervailing duties, and other trade-restrictive measures consistent with the final determination can be implemented on imported goods. Generally, the DOC should carry out a sunset review after five years. If it is determined that dumping, subsidizing, and corresponding material injuries do not exist, trade-restrictive measures are terminated. However, relative measures will be extended if the USITC and the DOC both determine that there is a possibility of continued dumping, subsidizing, and injury. In addition, administrative reviews and changed circumstances reviews are carried out in response to review requests during trade investigations.

2.2. Theoretical analysis

In this paper, a simplified analytical framework based on new trade theory (Krugman, 1979, 1980) and new-new trade theory (Melitz, 2003) is introduced to theoretically explore the impact of the EU and

[1] PV cells and modules fall under harmonized system (HS) code 854140. Given the variations in the HS high-level classification rules for PV cells and modules among different countries, this paper provides a preliminary estimation of the trade growth rate for PV cells and modules in the policy background section, which is based solely on the trade statistics published by the United Nations Commodity Trade Statistics Database (UN Comtrade).

[2] The author has derived conclusions on the basis of data from the Ministry of Commerce's statistics.

USA antidumping and countervailing measures on Chinese PV industry exports and the entry of new firms. Identifying the mechanisms by which expanding the domestic market and improving technological levels are effective in response to trade barriers is an important task for assessing the effects of domestic and technological innovation in promoting international circulation (see Appendix A).

3. Research design

3.1. Model specification

3.1.1. Export resilience research model

In this paper, the impact of ACPs initiated by the USA in 2011 and by the EU in 2012 on Chinese PV trade are empirically explored, and crystalline silicon PV cells and modules are the study objects. To accurately determine the policy effects of ACPs, policies during the same period that may have interfered (such as via economic environment adjustments and nontargeted policy perturbations) with PV trade trends are identified and excluded; notably, the DID method, which focuses on the difference between the PV trade volume in the treatment group and the control group before and after trade barriers, is implemented, and an in-depth analysis of the effects of ACPs is performed.

Chinese exports of PV cells and modules to the USA and EU during the study period are used to form the treatment group, and those to the rest of the world are used to form the control group. The difference between the treatment and control economies lies in the former's imposition of ACPs on Chinese PV exports.[3] Owing to the short intervals between the filing, preliminary ruling, and final ruling stages in the ACP process, with the overall process being less than one year, and though this paper is based on annual trade statistics, focus is placed on two time points, the filing and final ruling periods, which have relatively long intervals and no overlapping years. Moreover, considering that the EU's ACP against China was terminated in September 2018 and that the U.S. has repeatedly restricted Chinese PV product exports through trade wars and Section 337 investigations since 2018, as well as the outbreak of the COVID-19 pandemic in 2020, these policies and external shocks are not within the scope of our study. Therefore, the impact of ACPs against China within five year after the final decision was considered. The specific setting form of the DID model is as follows:

$$Y_{it} = \alpha + \beta_1 Treat_i \times Poltime_{it}^1 + \lambda X_{it} + \mu_i + \theta_t + \varepsilon_{it} \quad (1)$$

$$Y_{it} = \alpha + \beta_2 Treat_i \times Poltime_{it}^2 + \beta_3 Treat_i \times Poltime_{it}^3 + \lambda X_{it} + \mu_i + \theta_t + \varepsilon_{it} \quad (2)$$

$$Poltime_{it}^1 = \begin{cases} 1, & t \in [t_1, t_2 + 5] \\ 0, & Others \end{cases}$$
$$Poltime_{it}^2 = \begin{cases} 1, & t \in [t_1, t_2) \\ 0, & Others \end{cases} \quad (3)$$
$$Poltime_{it}^3 = \begin{cases} 1, & t \in [t_2, t_2 + 5] \\ 0, & Others \end{cases}$$

In the above equations, Y_{it} represents the export value of PV cells and modules from China to country or region i in year t. $Treat_i$ is a binary treatment variable that equals 1 if country or region i belongs to the treatment group and 0 otherwise. $Poltime_{it}^m$ is a binary policy timing variable ($m = 1, 2, 3$). Notably, $Poltime_{it}^1$ equals 1 if year t is within five years of the filing of the investigation and 0 otherwise; $Poltime_{it}^2$ equals 1 if year t is between the filing of the investigation and the final ruling and 0 otherwise; and $Poltime_{it}^3$ equals 1 if year t is within five years after the final ruling and 0 otherwise. i represents countries or regions that trade in PV cells and modules with China. t is the time variable, where t_1 represents the year when the EU and the USA initiated a corresponding ACP against Chinese PV exports, t_2 represents the year when the final ruling was made, and t_2+5 represents the period of five years after the final ruling. The investigation filing years for the USA and the EU were 2011 and 2012, respectively, and the final ruling years were 2012 and 2013, respectively.[4] The rationale for setting the policy variable on the basis of the initiation and final ruling time points is that while the EU and the USA ACPs against China's PV industry encompass three pivotal moments, namely, filing, the preliminary ruling, and the final ruling, there is a marked brevity in the intervals between these time points, resulting in overlap in terms of years. For example, the USA initiated its investigation against China's PV products in November 2011, with preliminary rulings occurring on March, 2012, followed by a final affirmative ruling on November 7, 2012. Similarly, the EU launched its ACP investigations against China in September and November 2012, with the preliminary antidumping case ruling in June 2013, culminating in a final affirmative ruling on December 5, 2013. Given that annual trade data are used in this study, it is most appropriate to select the distinct time points of investigation initiation and final ruling for conducting the research. X_{it} represents control variables that may affect Chinese exports of PV cells and modules to various countries, such as the economic development level, population size, annual new solar installation capacity, exchange rate, and other variables. μ_i is the country or region fixed effect. θ_t is the year fixed effect, which is used to control for country or region characteristics that do not change over time and other potential time-varying disturbances. α is a constant term, and ε_{it} is the random error term. To address the effects of heteroskedasticity and autocorrelation, clustered standard errors are determined at the country level.

3.1.2. Industrial resilience research model

Furthermore, to examine the resilience of China's PV industry and the impact of the EU and USA trade barriers on the Chinese PV industry, national business firm registration data and patent application data from 2009–2018 are collected, and the records of PV-related firms and patent applications are screened. On this basis, statistics on the registration of industrial and commercial firms, as well as the number of PV-related firms and patent applications by country and year, are obtained, with the new registrations in the PV industry and the number of PV-related patents in each country forming the treatment group and the number of registrations in non-PV industries and the number of non-PV-related patents in each country forming the control group. A standard DID model is employed to test the impact of ACPs on new firm registrations and PV-related patent applications in China. Specifically, the following DID model is constructed:

$$Y_{ijt} = \alpha + \gamma Indus_{ij} \times Time_t + \lambda X_{it} + \mu_{ij} + \theta_t + \mu_{ij} * \theta_t + \varepsilon_{it} \quad (4)$$

where Y_{ijt} represents the number of firm registrations or patent applications related to industry j in country i in year t. $Indus_{ij}$ is the policy treatment variable, which equals 1 if industry j is within the PV industry and 0 otherwise. $Time_t$ is the policy timing variable, which equals 1 for years t after 2012 and 0 for earlier years.[5] X_{it} is a set of variables that may affect PV industry development in a particular

[3] Information on global antidumping, countervailing and safeguarding cases related to PV products was obtained from the Temporary Trade Barriers Database developed by the World Bank, which can be accessed at the following address: https://www.worldbank.org/en/data/interactive/2021/03/02/temporary-trade-barriers-database.

[4] Given that the United States and the European Union made final rulings against Chinese products in November 2012 and December 2013, respectively, the final ruling years for the United States and European Union countries are set as 2013 and 2014 in the regression model.

[5] Given that the imposition of antidumping and countervailing duties by either the United States or the European Union began to impact the development of the Chinese PV industry, this paper designates the year 2012, when the USA first made the final ruling against China, as the policy shock time for studying industrial resilience. A standard DID model is employed.

country, including economic development level, topographical conditions, and hours of sunlight, among other variables. μ_{ij}, θ_t and $\mu_{ij} * \theta_t$ represent the country fixed effects, year fixed effects, and interaction fixed effects between country and year, respectively, to control for the impact of unobservable variables on the model estimation. α is a constant term, and ε_{it} is a random error term. Standard errors are clustered at the country level.

3.2. Variables and data

To compare the effects of the EU and the USA ACPs against China, the year 2009, which precedes the initiation of the EU and the USA ACPs, is selected as the starting year for the sample observation period. Additionally, to exclude various factors unrelated to this study, the year 2019 is set as the last year of the sample observation period.

Export the data for the PV cells and modules. The core dependent variable in this paper, Y_{it}, is the logarithm of the total export value of PV cells and modules from China to country or region i in year t (denoted as Logval_pv). Additionally, the export volume (denoted as Logquan_pv) is employed as the dependent variable for the robustness test. The variables Logval pv and Logquan pv are calculated by taking the logarithm of the original data directly, and the same process is used for zero values for these variables. Given the availability of trade data, the earliest accessible trade data for Chinese PV cells and modules were from 2009. To exclude various factors unrelated to this study, the year 2019 is set as the end year for the trade data analysis. The PV trade statistics from 2009–2019 are sourced from the Trade Map database.[6] The data are retrieved on the basis of the 8-digit HS codes "8541.4020" and "8541.4090" for PV products exported from China,[7] and the export trade value and volume of PV cells and modules to various countries in each year are aggregated. The export value is measured in current US dollars, and the export volume is measured in kilograms.

PV industry new firm data. Over 39.4 million Chinese industrial and commercial firm registration data points, ranging from January 1, 2009, to December 31, 2018, were obtained from the Qichacha website. Compared to the datasets circulating online that provide unclear registration information for firms, individual industries, and commercial entities, this dataset includes more complete firm registration information that can be used to eliminate the interference associated with new individual industrial and commercial entities in empirical research. It is possible to count the number of new firms in each industry category in each country by year during the sample period based on the information of "Establishment Date", "District County", and "Industry". Although "Industry" in the firm registration records only provides the information on the industry category to which the firm belongs, it is still possible to identify firms in the PV industry chain based on the "Business Scope" information via regular expressions. Specifically, based on key product information in the PV industry chain, over 540,000 newly registered firms in the PV industry chain during the sample period are identified via filtering according to whether "Business Scope" includes any of 18 keywords.[8] Subsequently, the number of newly registered PV firms in each county (denoted as entnum) each year was counted.

PV-related patent application data Following the methods of Nelson et al. (2022) and Liu et al. (2022), the number of patents is used as an indicator of corporate innovation (Wang et al., 2024). A total of 19.3 million patent application records were collected from the China National Intellectual Property Administration's patent database from 2009–2018. On the basis of the address information of the patent data, each patent application was matched with county-level administrative units. Since the patent application information lacks industry-specific details, all patents are categorized solely into PV-related and non-PV-related patents. There are two methods of classification: first, based on whether the patent's abstract contains any of the aforementioned 18 keywords (denoted as patent1), and second, referring to Binz et al. (2017), based on patents with classification codes falling within the set[9] are determined to be PV-related patent s(denoted as patent2). On the basis of the above two methods, the numbers of PV-related and non-PV-related patent applications in each county annually were determined.

The DID variable is the core independent variable. When exploring the export resilience of the Chinese PV industry, exports of Chinese cells and modules to the USA and the 28 EU countries[10] were used as the treatment group, and exports to other countries and regions were used as the control group; consequently, the DID variable $Treat_i \times Poltime_{it}^m$ was constructed. When examining the resilience of the PV industry in China, the numbers of PV industries and PV-related patents in each country were used to establish the treatment group, and the numbers of non-PV industries and patents in each country were used to establish the control group; consequently, the DID variable $Indus_{ij} \times Time_t$ was constructed. Detailed information on global antidumping, countervailing, and safeguarding measures within the sample period was obtained from the Temporary Trade Barriers Database developed by the World Bank (Signoret et al., 2020), and statistical data was collected from the Ministry of Commerce of China for corroboration.[11]

The control variables in this paper are set as follows.

A country's level of economic development and population size can have a significant effect on its level of trade and energy demand. Consequently, this paper includes the logarithm of the gross domestic product (log_GDP) and the logarithm of the population size of each country (log_population) in each year as control variables. The annual average exchange rate of various currencies in relation to the RMB (EXR) is also incorporated into the regression. The data are obtained from the World Bank's World Development Indicators (WDI) database[12] and then logged for inclusion in the model. Additionally, the annual new PV installation capacity data for each country (log_PV capacity) are obtained from the International Renewable Energy Agency (IRENA).[13] These data are used to reflect the PV application demand and log of this demand.

The market environment is another crucial factor affecting trade. In this paper, two indicators are used to characterize the market

[6] Trade Map, International Trade Centre, www.trademap.org developed by the International Trade Centre (ITC), a joint agency run by the World Trade Organization (WTO) and the United Nations Conference on Trade and Development (UNCTAD).

[7] Given the change in HS codes in 2012, a verification process was conducted to ensure the consistency of the HS 8-digit codes for the PV cells and modules studied within the sample period. It was confirmed that there was no adjustment in the coding throughout the research period.

[8] "photovoltaic", "solar energy", "photovoltaic modules", "solar cells", "crystalline silicon", "monocrystalline silicon", "polycrystalline silicon", "silicon wafers", "silver paste", "PET substrate", "fluorine film", "EVA", "inverter", "battery", "junction box", "EPC", "Hanergy" and "Tianwei".

[9] C01B, C23C, C30B, B28D, C08G, B01J, C07C, C07D, C08F, C08L, C25B, C01G, C08K, B22D, C07F, B23Q, B22F, B23P, B82B, B29B, B23K, H01B, H01L, H01G, G01B, G01R, G05F, B32B, B05D, G02F, G02B, H05K, H01J, G09G, H01H, C08C, G08C, H03K, C03C, C09K, G01N, C25D, G05B, C09D, G05D, H02P, G01J, H01F, A61B, G01D, H01S, C09J, H04Q, B05B, B65B, C09B, G03G, G03F, H01Q, C03B, B01F, G03C, B01D, B05C, B41M, C09C, F16M, B23B, C07K, C12M.

[10] As Croatia formally joined the EU in July 2013, the country was treated as follows: Croatia was not part of the treatment group when the EU initiated ACPs regarding China in 2012 and was included in the treatment group in 2013 when a final decision was reached.

[11] Trade investigation case data and related information were obtained from the China Trade Remedies Information Database of the Ministry of Commerce, PRC.

[12] Data can be obtained from the following address: https://databank.worldbank.org/reports.aspx?source=World-Development-Indicators.

[13] Data can be obtained from the following address: https://www.irena.org/Data/Downloads/ IRENASTAT.

Table 1
Summary statistics.

Variable	N	Mean	Sd	Min	Max
Dependent variables					
value_pv(1000$)	2276	76 260.731	4.03e+05	0	7.62e+06
Logval_pv	2030	7.316	3.116	0	15.846
entnum	949 323	45.827	356.75	1	59 377
patent1	51 089	393.417	1849.007	0	85 566
patent2	51 089	459.821	1885.159	0	85 566
Independent variables					
$Treat_i \times Poltime_{it}^1$	2276	0.101	0.302	0	1
$Treat_i \times Poltime_{it}^2$	2276	0.025	0.156	0	1
$Treat_i \times Poltime_{it}^3$	2276	0.076	0.266	0	1
$Indus_{ij} \times Time_t$	949 323	0.02	0.142	0	1
$Indus_{ij} \times Time_t$	51 089	0.343	0.475	0	1
log_GDP(1000$)	2150	17.246	2.313	10.770	23.745
log_population(1000)	2210	8.483	2.246	2.326	14.140
log_PV capacity(MW)	2262	1.498	2.146	−2.040	9.323
FTA	2238	0.098	0.298	0	1
ACP case	2276	1.119	2.596	0	26
EXR	2116	2.579	3.704	0	24.459
Average nighttime light	56 166	1.824	6.006	0	6.54E+01
DEM(m)	56 166	589.051	767.976	2.297	5145.42
daylight hours(h)	56 166	1982.91	519.38	0	3548.565

environment in which Chinese PV exports are located. First, data on regional trade agreements between countries are obtained from the CEPII database, and a dummy variable (FTA) related to the existence of a free trade agreement between China and its trading partners is established; it is coded as 1 if an agreement exists and 0 otherwise.[14] Second, the number of antidumping and countervailing cases involving China each year was determined; information on global antidumping and countervailing cases against China from the World Bank's Temporary Trade Barriers Database within the sample period was obtained[15] and summed to establish the ACP case indicator.

The level of regional economic development, topographical conditions, and solar irradiation time are all factors that influence the regional PV development potential. Therefore, the average nighttime light brightness from 2006–2008 in each country, the interaction term between the digital elevation model (DEM) and year, and the annual solar irradiation hours are used as control variables in Eq. (4). The average nighttime light brightness data from 2006–2008 for each country are from the global NPP-VIIRS nighttime light database (2000–2022) developed by Chen et al. (2021), the digital elevation model for each country is derived from the Copernicus Digital Elevation Model (COP-DEM), and the annual solar irradiation data for each country are from the China Surface Climate Data Daily Dataset.

3.3. Descriptive statistics

Table 1 presents the descriptive results for the model variables.

4. Empirical results and analysis

4.1. Benchmark regression results

4.1.1. Average treatment effect

In this paper, a DID model is constructed to examine the impact of the EU and the USA antidumping and antisubsidy cases on Chinese PV product exports. Table 2 reports the benchmark regression results. Columns (1) and (2) report the results when only the impact of the filing investigation shock is considered. Column (3) reports the results distinguished based on the filing investigation and the final ruling decision at two time points. First, the model includes only the policy variable for regression. On this basis, other control variables are included in the regression, with year fixed effects and country/regional fixed effects considered. According to the results in the first two columns in Table 2, compared with the trade situation between China and economies that have not initiated ACPs, the trade barriers between China and Europe and the USA has had a significant negative impact on Chinese exports of PV cells and modules. Following the filing investigation shock, the value of PV cells and modules exported to Europe and the USA decreased by 87.5% ($1 - e^{-2.080}$), and this result is significant at the 1% level. The results in column (3) indicate that ACPs significantly reduced Chinese exports of PV products to Europe and the USA following the filing investigation stage, with an average decrease of 69.7% in export value. The final ruling further contributed to sharply reducing Chinese exports to Europe and the USA, with an average decrease of 92%. The regression coefficients are all significant at the 1% level. The above results indicate that the ACPs have had a very significant trade disruption effect, which has been evident since the start of the investigation and become more pronounced since the European and American authorities issued affirmative final rulings.

Table 2
Impact of the EU and USA ACPs on chinese PV product exports.

Variable	(1) Logval_pv	(2) Logval_pv	(3) Logval_pv
$Treat_i \times Poltime_{it}^1$	−2.329***	−2.080***	
	(0.399)	(0.310)	
$Treat_i \times Poltime_{it}^2$			−1.194***
			(0.322)
$Treat_i \times Poltime_{it}^3$			−2.527***
			(0.345)
Control	NO	YES	YES
Year	YES	YES	YES
Country/region	YES	YES	YES
Constant	7.595***	0.266	7.271
	(0.046)	(12.432)	(12.728)
adj. R2	0.790	0.825	0.829
N	2024	1858	1858

[a] Note: Numbers in parentheses in each column are cluster-robust standard errors; *, **, and *** denote significance levels of 10%, 5%, and 1%, respectively; the definitions in the later table are the same as here.

[14] Data can be obtained from the following address: http://www.cepii.fr/CEPII/en/bdd_modele/bdd_modele_item.asp?id=8.

[15] The number of PV product-related cases included in the total number of antidumping and countervailing cases involving China in each country each year.

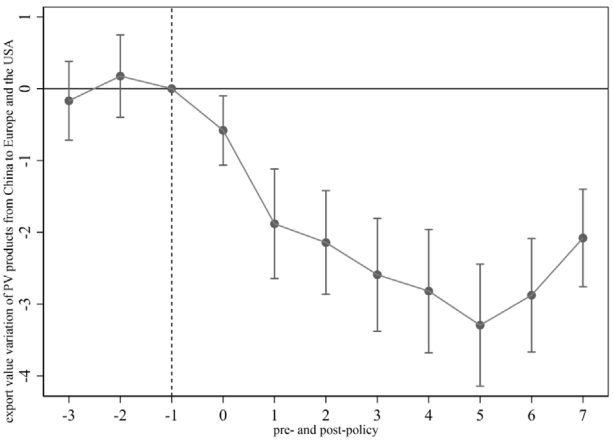


Fig. 1. Parallel trend test results for the PV cell and module export values.

4.1.2. Analysis of dynamic effects

A crucial prerequisite for causal identification based on the DID model is to satisfy the parallel trends assumption, which requires that the treatment and control groups show the same development trend before a policy shock. In this context, the following model is constructed to test the parallel trends between the treatment and control groups on the basis of the approach of Beck et al. (2010) and Sun and Min (2024).

$$Y_{it} = \alpha + \gamma_1 D_{it}^{-3} + \gamma_2 D_{it}^{-2} + \cdots + \gamma_{13} D_{it}^{+7} + \lambda X_{it} + \mu_i + \theta_t + \varepsilon_{it} \quad (5)$$

where D_{it}^{-j} is $Treat_i \times Poltime_{it}^{-j}$, which is set as 1 if country or region i in the treatment group is active j years prior to the initiation of the ACPs and 0 otherwise. Similarly, D_{it}^{+j} is $Treat_i \times Poltime_{it}^{+j}$ and is set to 1 if country or region i in the treatment group is active in the j years following an investigation and 0 otherwise. The regression coefficients of the variables represent the difference in PV trade between China and the treatment group and China and the control group countries or regions in various years. If the regression coefficients are not significant before a policy shock, the parallel trends assumption is validated. The meanings of the other variables in the model are the same as those in Eq. (1).

Fig. 1 presents parallel trend test results, where the gray dots represent the regression coefficients of the variable $D_{it}^{\pm j}$ in Eq. (5) and the vertical lines passing through the gray dots indicate the 95% confidence intervals of the coefficients. Fig. 1 shows that prior to the ACPs implementation, Chinese exports of PV cells and modules to the treatment and control groups showed the same time trend, satisfying the parallel trends assumption and simultaneously verifying the rationality of using the DID model.

Considering that existing DID mode-based studies have suggested that pretreatment trend tests cannot serve as effective empirical evidence for the parallel trends assumption (Roth, 2022) and may lead to biased estimation results, the approach of Rambachan and Roth (2023) is used by supplementing the traditional parallel trends test with a sensitivity analysis using relative deviation constraints and smoothness constraints to examine the robustness of the model results. Specifically, following the method of Biasi and Sarsons (2021), the maximum deviation from the parallel trend (Mbar) is set to one standard error, and the confidence interval of the posttreatment point estimates is analyzed accordingly. As shown in Fig. 2, under the relative deviation constraint, the effect of the ACPs is still significantly negative at the 95% confidence interval; under the smoothness constraint, when the pretreatment trend deviation is less than 50%, the effect of the ACPs remains significantly negative. The aforementioned results indicate that even with some deviation from the parallel trend, ACPs still have a significant negative effect on China's photovoltaic exports.

Notably, the analysis method based on the event study model, expressed in Eq. (5), can not only be used to test the preintervention parallel trends but also provide information about the corresponding dynamic effects (Miller, 2023). As shown in Fig. 1, compared with the year just before the filing of the ACPs, the regression coefficients of the variable D_{it}^{-j} did not pass the significance test in the years before the policy shock; in the year the investigation was initiated and for several years thereafter, the regression coefficients were all significantly less than 0. Notably, after the investigation, Chinese exports of PV cells and modules to the treatment and control group countries or regions showed a trend of first falling sharply and then gradually recovering. This result indicates that the short-term and long-term impacts of the EU and the U.S. ACPs were heterogeneous. In the long term, the negative impact of the EU and the US ACPs is not sustainable, implying that as time progresses, further cost reduction[16] will enable Chinese PV product export prices to regain a price advantage even after the imposition of antidumping and countervailing duties, thereby increasing competitiveness in the international market.

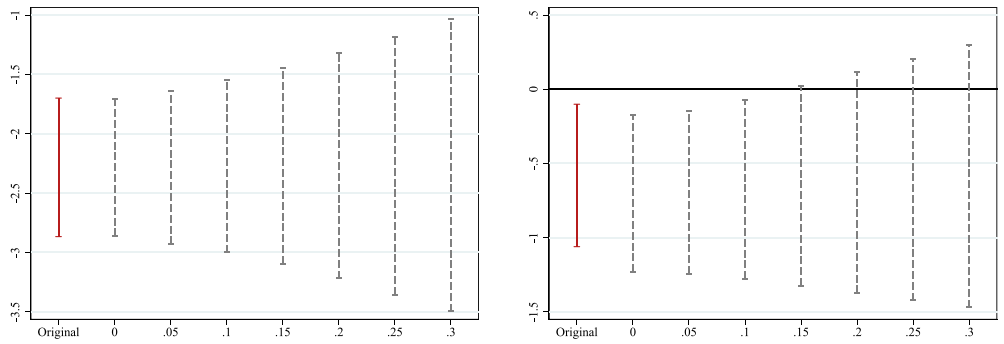
4.2. Robustness test

4.2.1. Bacon decomposition

In the existing theoretical econometric literature, it has been shown that in staggered DID models, the DID estimator using two-way fixed effects (TWEEs) may be biased due to the presence of heterogeneous treatment effects (Borusyak et al., 2024; Sun and Abraham, 2021). Goodman-Bacon (2021) argued that staggered DID estimators should be obtained as a weighted average of three sets of 2 × 2 DID estimators: the first set is based on comparing all treatment groups (including early and late treatment groups) with the never-treated group, the second set is based on comparing early treatment groups with late treatment groups, and the third set is based on comparing late treatment groups with early treatment groups. The first and second sets can be considered valid controls, whereas the third set may lead to biased DID estimators under the influence of heterogeneous treatment effects, especially when the weight or coefficient of this group is quite large.

In this paper, the decomposition method of Goodman-Bacon (2021) is used to assess the bias in the TWFE estimator in benchmark regression. Table 3 and Fig. 3 report the decomposition results, where the DID estimate of −2.080 obtained from the TWFE model in benchmark regression is a weighted sum from the following groups: "Never_v_Timing" represents the 2 × 2 DID estimates (the policy effects obtained from the treatment group compared with the control group that never received treatment) when all treatment groups are compared with the group that has never been treated, and "Timing_groups" represents the 2 × 2 DID estimates obtained by comparing early treatment groups with late treatment groups, and vice versa (the policy effects obtained from the second and third groups compared with their respective control groups). The results in Table 3 show that approximately 91.9% of the negative effect of the EU and the USA ACPs on Chinese exports for countries in the treatment group is unbiased according to 2 × 2 DID estimates. Since this type of sample constitutes the majority of the overall sample in this study, it is reasonable to believe that the DID estimate obtained via the TWFE model has minimal bias. Moreover, the decomposition results show that the sum of the DID coefficients obtained from the "Timing_groups" comparison is positive, indicating that the DID estimate obtained via benchmark regression may underestimate the effects of the EU and the USA ACPs.

[16] According to IRENA, from 2010–2021, the installed cost of PV power generation was reduced by 82.17%, and the levelized cost of PV power generation was reduced by 88.49%.



(a) Relative deviation limit (b) Smooth limitation

Fig. 2. Parallel trend sensitivity test.

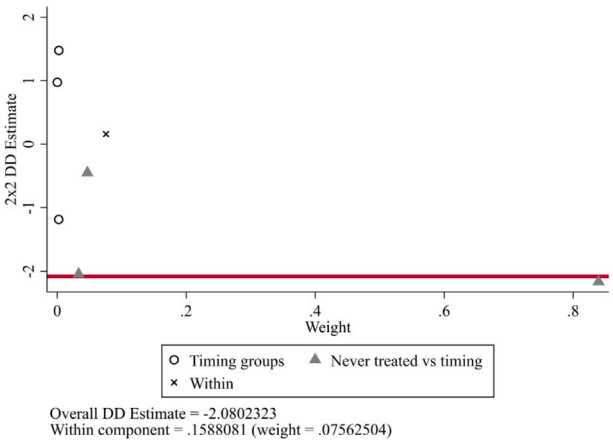


Fig. 3. Parallel trend test results for the PV cell and module export values.

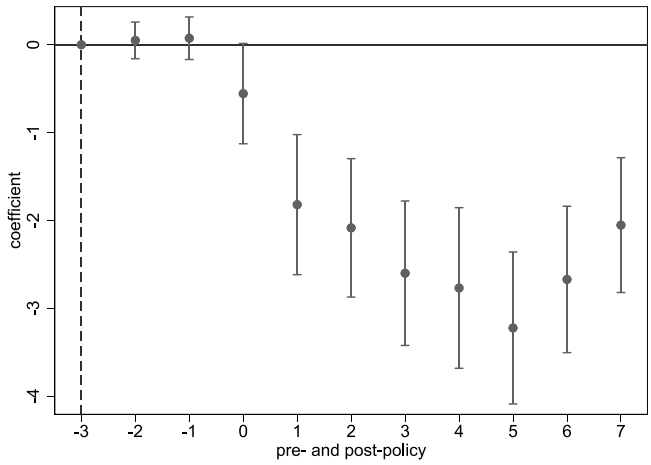


Fig. 4. Parallel trend test results for the PV cell and module export values.

Table 3
Bacon decomposition results.

Item	Coefficient	Weight
Never_v_Timing	−2.074	0.919
Timing_groups	0.224	0.005
Within	0.158	0.076

[a] Never_v_Timing in this table is the 2 × 2DID estimate for the treatment group compared to that for the never treated group, and Timing_groups represents the 2 × 2 DID estimates obtained by comparing early treatment groups with late treatment groups, and vice versa.

4.2.2. Staggered DID estimator

In this paper, the bias of the TWFE estimator in the benchmark model is confirmed to be minimal through Bacon decomposition. However, the decomposition method proposed by Goodman-Bacon (2021) offers an intuitive approach to effectively test the bias of static effects in the DID model. Nonetheless, the robustness of the estimator in the dynamic DID model is challenging to verify via this method. Therefore, the two-stage DID estimator proposed by Gardner (2022) is used to replace the TWFE estimator in the benchmark model, and another regression analysis is performed. This approach, from the perspective of dynamic modeling, facilitates assessments of the policy effects of the EU and the USA ACPs.

The robustness estimation results are presented in Fig. 4, where the horizontal axis represents the robustness estimation of the policy effect relative to the *i*th year of the sample. Prior to the implementation of ACPs for Chinese PV products by the EU and the USA, the regression coefficient of the robustness estimate does not significantly differ from zero, indicating that the treatment group and the control group had similar development trends, thus satisfying the parallel trend assumption. In the first year following the implementation of ACPs, Chinese exports of PV products to the treated countries significantly decreased, and this negative impact has persisted over time. The negative impact initially intensified and then diminished within the sample period, with the results being statistically significant at the 1% level. This suggests that ACPs of Europe and the United States against China have caused substantial damage to PV trade, and Chinese PV exports initially declined sharply after the policy's onset and then gradually decreased, indicating a trend of trade recovery toward stability. On the basis of the robustness estimation results, it is reasonable to suggest that the TWFE estimator in the benchmark regression provides a robust estimation of the policy effects of European and American ACPs.

Moreover, a robustness test of "placebo test", "replace dependent variable and eliminate outliers", "replacement of control group", "replacement of regression model", "refinement of control variables" and "differential impact of ACPs measures for the US and for the EU" was conducted, and the results show that the conclusions of this paper remain valid; see Appendix B for details.

5. Further discussion: Industrial resilience

5.1. Discussion from the perspective of newly registered firms in the PV industry

In this paper, the impact of ACPs of the EU and the USA on the number of newly registered firms in the Chinese PV industry is explored using the number of newly registered firms by county and industry as

Table 4
Impact on the new entrants in the chinese PV industry.

Variable	(1) entnum	(2) entnum	(3) entnum	(4) entnum	(5) entnum
Did	0.376***	0.375***	0.375***	0.331***	−0.107***
	(0.031)	(0.034)	(0.022)	(0.018)	(0.017)
Control	NO	YES	YES	YES	YES
Year	YES	YES	YES	YES	YES
County	YES	YES	YES	YES	YES
County-year	NO	NO	NO	YES	YES
Clustered standard errors	County–industry	County–industry	County	County	County
Constant	5.926***	5.870***	5.870***	7.714***	5.904***
	(0.000)	(0.065)	(0.088)	(2.015)	(0.697)
pseudo R2	0.932	0.933	0.933	0.950	0.948
N	832 090	832 090	832 090	832 053	358 845

[a] The values in parentheses in columns (1) and (2) represent robust standard errors clustered at the county–industry level, while those in parentheses in columns (3) to (5) represent robust standard errors clustered at the county level. *, **, and *** denote significance levels of 10%, 5%, and 1%, respectively.

the dependent variable on the basis of Eq. (4). Notably, the number of firms is a typical count variable with many zero values, and this distribution characteristic poses a challenge to regression analysis (Cohn et al., 2022). Therefore, Poisson regression is used to establish Eq. (4).

The regression results are presented in Table 4. Columns (1) and (2) display robust standard errors clustered at the county–industry level, whereas columns (3) to (5) show the clustering results for the standard errors at the county level. Columns (4) and (5), which are based on the inclusion of control variables, reflect further control for county-year fixed effects to eliminate the influence of confounding factors specific to each county in certain years. Columns (1) to (4) of Table 4 show that ACPs do not significantly reduce the number of newly registered firms in the Chinese PV industry; in contrast, they increase the number of new registrations at the 1% level of significance.

To identify the reasons for these unexpected results, the event study method is used to explore the dynamic effects of ACPs of the EU and the USA on the number of newly registered firms in the Chinese PV industry. The results are shown in Fig. 5 and indicate that ACPs indeed had a significant negative effect on the development of the Chinese PV industry in 2012 and 2013, the years immediately following the introduction of trade barriers. When the sample period is truncated to 2009–2013, this negative impact leads to a 10.7% decrease in newly registered firms in the Chinese PV industry (as shown in column (5) of Table 4). Encouragingly, after two years of difficulty in 2012 and 2013, the Chinese PV industry began to recover in 2014, with the number of newly registered firms in each year being significantly greater than that in the counterfactual control group and the recovery period exceeding the negative impact, ultimately showing a positive static effect. This result is consistent with the spatiotemporal trend of Chinese PV product exports under the influence of the ACPs with the difference being that the recovery speed of the Chinese domestic PV industry is comparatively faster. In summary, although the ACPs of the EU and the USA had a statistically and economically significant negative impact on the Chinese PV industry in the short term, the industry has displayed resilience and quickly recovered from external shocks, showing a development trend better than that before the ACPs were established.

To determine why the number of newly registered PV firms in China followed a "V-shaped" recovery trend in terms of PV exports, statistics on the export value of Chinese PV cells and modules from 2009–2021 were obtained, and the cumulative installed PV capacity, cumulative grid-connected PV capacity, and new PV installation capacity in China from 2001–2022 were collected. The relevant data are presented in Fig. 6. From 2009 to 2011, the export value of Chinese PV cells and modules grew rapidly, whereas the domestic installed PV capacity remained relatively low, indicating that Chinese PV products were primarily directed toward overseas markets. However, after the implementation of ACPs of the EU and the USA, the export value of Chinese PV cells and modules experienced a steep reduction in 2012

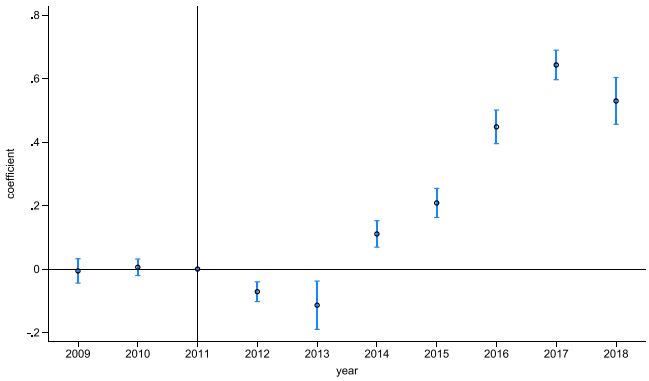


Fig. 5. Analysis of the dynamics of new PV industry entrants.

and remained relatively stable from 2012–2018, without showing a recovery trend. In contrast, Fig. 5 indicates that the number of newly registered PV firms in China had already returned to pre-ACP levels by 2014.

To quantitatively link domestic market demand to "V-shaped recovery", following Li et al. (2019), we employ a heterogeneity analysis approach to investigate whether regions with greater numbers of installed PV power stations experienced a more substantial increase in PV firms after the implementation of the ACPs of the EU and the USA than other regions did. Specifically, we use the 2010–2022 spatial extent dataset of PV power plants in China from the National Platform for Earth System Science Data Sharing Service - Yangtze River Delta Science Data Centre (Chen et al., 2024). The data are based on the Sentinel-2 and Landsat datasets, and the advanced semantic segmentation model TransUNet is used to extract the PV power plants. The continuous change detection and classification (CCDC) algorithm is used to estimate the installation date of each PV power plant. The attributes of the data include the power plant installation date, area and main type of occupied land. On the basis of the vector graphics of the PV plants, we employ ArcGIS to calculate the installed area of the PV power stations in each county from 2010–2018. Furthermore, we construct the following model to examine the relationship between local market demand and the recovery of PV enterprises:

$$Y_{ijt} = \alpha + \beta PVarea_{it} + \gamma Indus_{ij} \times Time_t + \zeta Indus_{ij} \times Time_t \times PVarea_{it} + \lambda X_{it} + \mu_{ij} + \theta_t + \mu_{ij} * \theta_t + \epsilon_{it} \quad (6)$$

where $PVarea_{it}$ represents the installed PV area in county i in year t and the other variables maintain consistent with the definitions in Eq. (4). ζ is our coefficient of interest, which reflects the stimulative effect

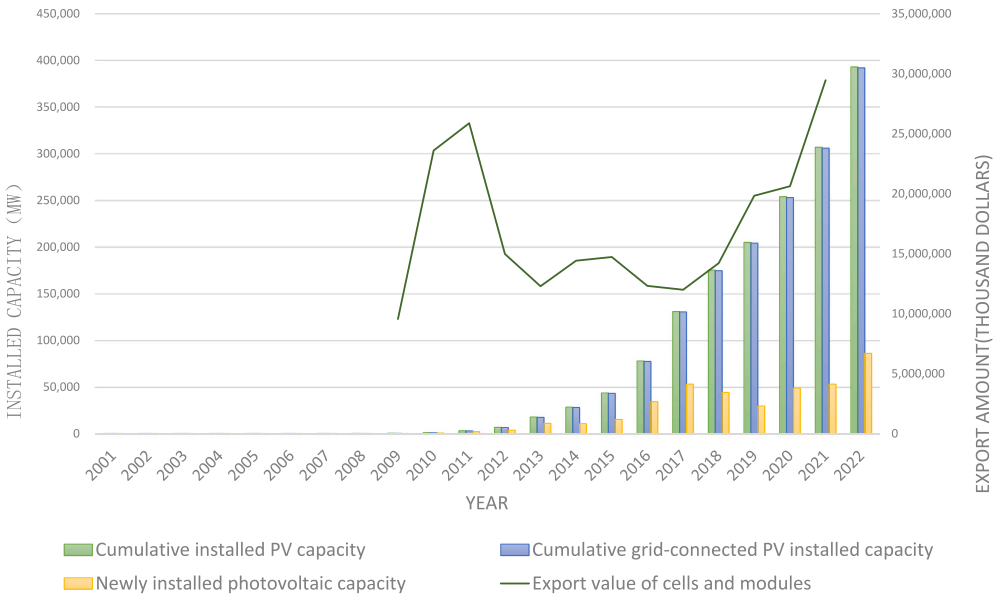


Fig. 6. Spatiotemporal evolution of the PV installation capacity in China.

Table 5
Domestic market demand and new entrants in the chinese PV industry.

Variable	(1) entnum	(2) entnum
Indus × Time	0.369***	0.368***
	(0.021)	(0.022)
Indus × Time × PVarea	0.125***	0.125***
	(0.045)	(0.046)
Control	No	YES
Year	YES	YES
County	YES	YES
Constant	5.977***	5.850***
	(0.001)	(0.091)
pseudo R2	0.935	0.936
N	758 309	758 309

of local PV market demand on the number of new PV enterprises. The regression results, as shown in Table 5, indicate that after the implementation of ACPs of the EU and the USA, for each additional thousand square kilometers of PV installation area in a county, the number of new PV firms in that county increased by 12.5%. This finding demonstrates the strong relationship between local market demand expansion and "V-shaped recovery".

Although this paper provides rigorous causal evidence, Fig. 6 shows that the domestic installed PV capacity in China began to exhibit a geometric growth trend starting in 2012, which led us to speculate that the domestic PV power generation in China provided a vast market for the entities in the PV industry impacted by the ACPs, effectively facilitating the rapid recovery of the Chinese PV industry.

The theoretical analysis in this paper indicates that R&D innovation and emphasis on technological advantages are important factors affecting industry recovery from external shocks. Therefore, on the basis of national patent application data from 2009–2018, PV-related patent applications in China are used as a quantitative indicator of innovative R&D intensity in the Chinese PV industry. We find that ACPs do not hinder innovative R&D in the Chinese PV industry (see Appendix C).

6. Conclusions and policy implications

For a considerable period, representative nontariff barriers such as antidumping and countervailing measures have severely impacted Chinese PV product exports and the development of the PV industry. Quantitatively evaluating the consequences of related trade policies is important for ensuring that Chinese green industry trade achieves secure, sustainable, resilient and high-quality development.

This study examines the short-term and long-term impacts of Europe and the USA trade barriers on Chinese PV product exports via a DID model and an event study method, with a focus on antidumping and countervailing policies implemented before and after 2011. It is based on Chinese PV trade data (HS 8-digit coding) from 2009–2019 and trade policy data from the World Bank's Temporary Trade Barriers Database. The study reveals that trade restriction policies have significantly negatively impacted Chinese PV exports, with adverse effects exhibiting temporal heterogeneity. In the short term, they led to a noticeable and substantial decrease in Chinese PV exports to Europe and the United States. In the long term, the negative impact on Chinese PV exports is not sustainable. The above conclusions are verified as reliable after multiple robustness tests. Furthermore, although antidumping and countervailing policies in the short term negatively impacted the entry of new firms into the Chinese PV market, the Chinese PV industry quickly recovered from the impact of external trade barriers and displayed a development trend better than that before trade barriers. This demonstrates good resilience in industrial development. Moreover, after the implementation of antidumping and countervailing policies by the EU and the USA, there was no significant decrease in the number of PV-related patent applications compared with non-PV-related patents in China, indicating that these trade barriers did not hinder innovation, research or development in the Chinese PV industry.

On the basis of the research findings in this paper, the following policy implications are derived.

First, trade wars are not only inadvisable but also unwinnable (Xia et al., 2019). From a historical perspective, while antidumping and countervailing measures imposed by Europe and the United States on China have impacted Chinese exports, they have not halted the rapid development of the Chinese PV industry. Through diligent internal efforts, China has gained a leading advantage and a dominant position in the global PV market, achieving an astonishing rebirth. However, the EU and the USA have lost a substantial amount of PV product imports. Faced with insufficient domestic production, they have found themselves in a passive situation where production activities are hindered and the market for clean energy applications is lacking. Trade blockades generally lead to a severe loss of foreign product supply, damaging the interests of the initiating countries.

Second, expand PV exports to emerging and domestic markets. The Chinese PV industry has faced severe pressure from Europe and the United States. The development strategy of the Chinese PV industry should be adjusted, and a diversified development approach should be adopted to achieve resilient high-quality growth. On the one hand, adapting to and seizing opportunities presented by globalization, expanding into overseas markets, and strengthening cooperation with economies via the Belt and Road Initiative, members of the RCEP, and BRICS countries in the fields of the PV industry and trade are essential tasks. The rational adjustment and spatial redistribution of the supply chain can reduce trade risk through the diversification of capacity and trade markets. On the other hand, China has become the world's largest carbon emitter (Li et al., 2023). Under the guidance of the dual circulation development strategy and the carbon peaking and carbon neutrality goals, fully leveraging the significant advantage of the vast Chinese domestic market is crucial. Similar patterns can be observed in China's electric vehicle (EV) sector, where comprehensive government support policies and robust domestic demand have fostered rapid industry development. As in the PV industry, Chinese EV manufacturers have achieved significant cost advantages through economies of scale and supply chain integration. However, both sectors face comparable trade challenges from Western markets, where trade restrictions may limit market access and technology exchange, potentially impeding global clean technology advancement and application. Recent empirical studies on the China–US trade war similarly find that the mechanism through which trade wars reduce firm innovation lies in the reductions in market expansion and innovation inputs (Kong et al., 2024). This, in turn, highlights the importance of expanding the domestic market and accelerating technological innovation in responding to trade barriers.

Third, the alignment of trade policies with climate commitments is crucial for the global energy transition. Our research demonstrates that trade barriers in renewable energy sectors significantly constrain the global deployment of clean power and the structural transformation of energy systems, thereby substantially impeding collective global efforts to address climate change. This issue becomes particularly critical in the context of China's commitment to reaching a carbon peak by 2030 and achieving carbon neutrality by 2060 (Cui et al., 2024; Liu et al., 2023); notably, international cooperation in renewable energy industries has emerged as an indispensable factor for realizing these ambitious climate goals. Consequently, nations worldwide should promote trade liberalization and deepen technological cooperation in renewable energy sectors within the framework of carbon reduction, avoiding trade barriers. This requires the establishment of specialized trade frameworks to facilitate clean energy technologies and product flows, the design of supportive policies to encourage technology transfer and knowledge sharing between developed and developing countries, and the creation of multilateral mechanisms to coordinate carbon pricing and environmental standards. As shifts in American climate policy could affect global climate cooperation, other countries may need to strengthen international coordination to expand renewable energy markets and accelerate technological innovation (Sun et al., 2024). The development of clean energy technologies and markets could proceed through multilateral frameworks and bilateral partnerships, focusing on areas such as market creation, technology transfer, and joint R&D initiatives.

CRediT authorship contribution statement

Mudan Wang: Writing – original draft, Formal analysis, Data curation. **Zeyu Xie:** Writing – original draft, Methodology, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Theoretical analysis

To simplify the analysis, the paper regards the EU and the USA as one economic entity, and China as another.

Consumer Preferences and Demand: The consumer characteristics are identical to those in Krugman (1980). The preferences of consumers for agricultural products and industrial products are of the Cobb–Douglas type, and the preferences within industrial products are of constant elasticity of substitution (CES), meaning that the consumer's maximization problem is as follows:

$$\max U_j \equiv c_0^{(1-\phi)} \left(\sum_{i=1}^{N-1} c_{ij}^{\frac{\sigma-1}{\sigma}} + c_{Nj}^{\frac{\sigma-1}{\sigma}} \right)^{\phi \frac{\sigma}{\sigma-1}} \quad \text{s.t.} \sum_{i=0}^{N} p_{ij} c_{ij} = w L_j \tag{A.1}$$

where c_0 represents the agricultural product that serves as a general equivalent, homogeneous in nature and without any trade costs, indicating that the agricultural product market is characterized by perfect competition and identical cross-country prices. U_j denotes the level of consumer utility in economy j, c_{ij} signifies the consumption quantity of product i in economy j. It is assumed that the Nth good is electricity, so c_{Nj} represents the consumption of electricity in economy j. At an early time period t0, such as before 2007, the Nth good, electricity, was primarily generated from fossil fuel energy sources like thermal power generation. p_{ij} is the price for the corresponding product, w represents the wage rate for labor, L_j is the quantity of labor force in economy j. ϕ and σ are model parameters.

The production function for the agricultural product is given by $y_0 = L_0$, and the price of the agricultural product is normalized to 1. Applying the zero-profit condition $\pi = p_0 y_0 - w L_0 = 0$, it is straightforward to derive that $w = 1$. Based on this, the demand function can be derived from consumer utility maximization at a given income level:

$$c_{ij} = \left(\frac{p_{ij}}{P_j} \right)^{\sigma} \left(\frac{\phi L_j}{P_j} \right) \tag{A.2}$$

where P_j represents the aggregate price index for economy j.

The cost function for industrial goods is identical to that in Krugman (1980), where firms use only labor for production, incurring both a fixed cost f and a variable cost that depends on the firm's productivity φ. Thus, the labor input required for a firm to produce a quantity q is given by:

$$l(q) = f + \frac{q}{\varphi} \tag{A.3}$$

In a closed economy, where and there is into the market, the pricing and output decisions of firms can be derived from firms maximize profits and market free entry conditions:

$$p = \frac{\sigma}{\sigma - 1} \frac{1}{\varphi} \tag{A.4}$$

$$q = (\sigma - 1) \varphi f \tag{A.5}$$

In an open economy with iceberg transportation costs τ, the consumer preferences, production cost functions, and demand elasticity remain unchanged and are identical across both economies. Therefore, the domestic price of industrial goods in the home country continues to be determined by:

$$p = \frac{\sigma}{\sigma - 1} \frac{1}{\varphi} \tag{A.6}$$

The price for products exported to foreign countries is given by τp. Up to this point, all the settings and conclusions are consistent with Krugman (1980), and the relevant derivations can be referred to in the original text.

Prior to time t_0, the electricity products on the market were exclusively supplied by fossil fuel-based power generation. At some point after, specifically at time t1, PV power generation began to emerge. For the representative consumer, the utility derived from thermal power generation and PV power generation is equivalent[17] , meaning that thermal and PV power generation are perfect substitutes.

As a new product, the production cost function for PV products is as follows:

$$I(q) = f + \frac{q}{\varphi_s} + q^{\alpha}, \alpha > 1 \tag{A.7}$$

In comparison to the mature technology of fossil fuel-based power generation, PV industry incurs higher variable costs, denoted as q^{α}. Concurrently, PV production technology is not yet mature, and the market structure is unstable. Therefore, the setup in Melitz (2003), where firms have heterogeneous productivity and there is firm entry and exit behavior, better aligns with the characteristics of the emerging photovoltaic products. Drawing from Melitz (2003), this paper posits the following regarding the photovoltaic (PV) industry: 1. Prior to the entry of potential entrants into the PV industry, firms are homogeneous; 2. The number of potential entrants that may enter the PV industry is M, Potential entrants do not incur any sunk costs upon market entry; 3. After entering the market, potential entrants will randomly acquire an initial productivity level, denoted as φ_s, φ_s is uniformly distributed over the interval (0, φ_{max}) according to the probability density function $g(\varphi)$, let $G(\varphi)$ be the continuous cumulative distribution function of $g(\varphi)$. The scenario where $\varphi_{max} = \varphi$ implies that the maximum productivity attainable by PV firms is nearly, but never equal to, the productivity of firms in the fossil fuel-based power generation industry. 4. Once a firm enters the market, it will decide whether to remain in the market based on its productivity in each subsequent period, without being forced to exit the industry due to random negative shocks.

To encourage PV industry development, it is assumed that both China and Europe and the USA will provide a one-time subsidy f to firms entering PV industry. Additionally, Europe and the USA will impose an extra carbon tax τ_c on fossil fuel power generation, which is considered as a subsidy for PV power generation in this study. This paper considers the carbon tax on fossil fuel power generation. First, we consider the case without PV products trade. The marginal costs and average costs for firms within the PV industry in China (subscripted as e) and Europe and the USA (subscripted as w) are as follows:

$$MC_e = \frac{1}{\varphi_s} + \alpha q^{\alpha-1}, MC_w = \frac{1}{\varphi_s} + \alpha q^{\alpha-1} - \tau_c \tag{A.8}$$

$$AC_e = \frac{1}{\varphi_s} + q^{\alpha-1}, AC_w = \frac{1}{\varphi_s} + q^{\alpha-1} - \tau_c \tag{A.9}$$

Unlike the monopolistic competition market structure of the fossil fuel power generation market, the market for new products is more akin to a perfectly competitive market. That is, PV products are produced by multiple firms, each without market power and can only accept the price given by the market. To simplify the analysis, under the generality assumption, since consumers have a relatively low consumption of PV electricity in the initial stage, it will not impact the original market equilibrium, especially the equilibrium of the fossil fuel electricity market. That is, the firms of fossil fuel electricity maintain the price $p = \frac{\sigma}{\sigma-1}\frac{1}{\varphi}$ and output at time t_0. Given that PV electricity and fossil fuel electricity are perfect substitutes, firms in PV electricity sector will also choose their output levels under the given price p to maximize profits. By setting $\rho = \frac{\sigma}{\sigma-1}$, we can determine the optimal quantity of PV production for firms in China, Europe and the USA:

$$q_e = \left(\frac{\rho}{\alpha\varphi} - \frac{1}{\alpha\varphi_s}\right)^{\frac{1}{\alpha-1}}; \quad q_w = \left(\frac{\rho}{\alpha\varphi} - \frac{1}{\alpha\varphi_s} + \frac{\tau_c}{\alpha}\right)^{\frac{1}{\alpha-1}} \tag{A.10}$$

According to the zero-profit condition in a perfectly competitive market, the critical productivity levels φ_{se}^* and φ_{sw}^* for Chinese and Western PV producers, respectively, which ensure that their output is not zero, can be derived as follows:

$$\varphi_{se}^* = \frac{\varphi}{\rho}; \quad \varphi_{sw}^* = \frac{\varphi}{\rho + \varphi\tau_c} \tag{A.11}$$

Given the distribution function of productivity levels in the PV industry φ_s φ_{max}, in the absence of international trade in PV products, the number of firms in the Chinese and Western PV industries, denoted as N_e and N_w:

$$N_e = \begin{cases} 0 \cdots \cdots, \left| \varphi_{\max} < \frac{\varphi}{\rho} \right. \\ \left(1 - \frac{\varphi}{\varphi_{\max}\rho}\right) M, \left| \varphi_{\max} \geq \frac{\varphi}{\rho} \right. \end{cases} \tag{A.12}$$

$$N_e = \begin{cases} 0 \cdots \cdots, \left| \varphi_{\max} < \frac{\varphi}{\rho+\varphi\tau_c} \right. \\ \left(1 - \frac{\varphi}{\varphi_{\max}(\rho+\varphi\tau_c)}\right) M, \left| \varphi_{\max} \geq \frac{\varphi}{\rho+\varphi\tau_c} \right. \end{cases} \tag{A.13}$$

It can be observed that even if firms do not incur the sunk costs as outlined in Melitz (2003), the number of firms deciding to remain in the market is still zero when the maximum attainable productivity is less than the critical productivity level.

Next, consider that manufacturers in PV industry can also participate in international trade by paying iceberg trade cost τ, and that exports of Chinese PV products to the West can also be subsidized by Europe and the USA for PV products. Here, this paper focuses on analyzing the number of PV manufacturers in China and the number of PV producers participating in international trade. The marginal cost of Chinese PV manufacturers in domestic market and the marginal cost of exporting under an open economy are MC_{ed} and MC_{ex}, respectively:

$$MC_{ed} = \frac{1}{\varphi_s} + \alpha q^{\alpha-1}, MC_{ex} = \frac{1}{\varphi_s} + \alpha q^{\alpha-1} + \tau - \tau_c \tag{A.14}$$

Similarly, based on the zero-profit condition, the critical productivity φ_{ed}^* and φ_{ex}^* distributions of Chinese PV producers participating in the domestic market and in international trade can be obtained as:

$$\varphi_{ed}^* = \frac{\varphi}{\rho}; \quad \varphi_{ex}^* = \frac{\varphi}{\rho + \varphi\left(\tau_c - \tau\right)} \tag{A.15}$$

The current world international trade transport costs are already low, and the unit transport costs τ are often lower than the subsidies given by governments to emerging industries τ_c. This implies that $MC_{ed} > MC_{ex}$, the technological thresholds of Chinese PV manufacturers to participate in international trade are lower than that of their participation in the domestic market, and that more manufacturers participate in international trade. This is consistent with the fact that PV products produced in China before 2012 were mainly for export.

Further, consider the case of European and American tariffs on Chinese PV products, because this paper focuses on the EU and the USA ACPs on Chinese PV exports and the number of PV firms, but not concerned about the welfare effect of the tariffs; therefore, this paper will set tariffs in the form of iceberg cos. That is, China exports 1 unit of PV products to Europe and the USA, need to pay tariffs τ_{tax}, but the European and American governments do not get corresponding tax, tariffs, like the iceberg cost of trade, disappeared in transit.

Since tariffs and iceberg trade costs affect firms' decisions in the same way, similar to the analyses under an open economy, it can be concluded that, in the case of tariffs imposed by Europe and the USA on Chinese PV products, the critical productivity of Chinese PV producers participating in the domestic market and participating in international

[17] Admittedly, there may be a minority of consumers who, in pursuit of their spiritual needs, perceive green energy to offer higher utility; however, this does not represent the general preference of the representative consumer.

trade, namely $\varphi_{ed}^{*\prime}$ and $\varphi_{ex}^{*\prime}$, are distributed as:

$$\varphi_{ed}^{*} = \frac{\varphi}{\rho}; \quad \varphi_{ex}^{*} = \frac{\varphi}{\rho + \varphi\left(\tau_c - \tau - \tau_{\text{tax}}\right)} \tag{A.16}$$

The number of Chinese PV producers participating in the domestic market and participating in international trade, N_{ed} and N_{ex}, are as follows:

$$N_{ed} = \begin{cases} 0 \cdots \cdots, \left| \varphi_{\max} < \frac{\varphi}{\rho} \right. \\ \left(1 - \frac{\varphi}{\varphi_{\max}\rho}\right) M, \left| \varphi_{\max} \geq \frac{\varphi}{\rho} \right. \end{cases} \tag{A.17}$$

$$N_{ex} = \begin{cases} 0 \cdots \cdots \cdots \cdots, \left| \varphi_{\max} < \frac{\varphi}{\rho+\varphi(\tau_c-\tau-\tau_{\text{tax}})} \right. \\ \left(1 - \frac{\varphi}{\varphi_{\max}[\rho+\varphi(\tau_c-\tau-\tau_{\text{tax}})]}\right) M, \left| \varphi_{\max} \geq \frac{\varphi}{\rho+\varphi(\tau_c-\tau-\tau_{\text{tax}})} \right. \end{cases} \tag{A.18}$$

Further, the total number of Chinese PV companies is $N_e = \{N_{ed}, N_{ex}\}$.

Base on Eqs. (A.15), (A.16), (A.17), (A.18) and $\frac{\partial N_{ed}}{\partial \varphi_{\max}} > 0$, $\frac{\partial N_{ex}}{\partial \varphi_{\max}} > 0$, $\frac{\partial N_{ex}}{\partial \tau_c} > 0$, $\frac{\partial N_{ex}}{\partial \tau_{\text{tax}}} > 0$, the following inferences can be drawn:

1. In the presence of perfect substitutes in the market, even if potential market entrants in the PV industry can obtain an initial productivity without paying a silent cost, there still exists a critical productivity φ_s^*, such that manufacturers with productivity lower than φ_s^* will not put into production and when the upper limit of the feasible productivity of PV manufacturers $\varphi_{max} < \varphi_s^*$, none of the potential market entrants will put into production, i.e. the number of PV firms in the market is zero;
2. all Chinese PV manufacturers will participate in international trade when $\tau_c - \tau - \tau_{tax} > 0$, and only those with productivity higher than φ_{ex}^* will supply both domestic and international markets, and those with productivity between $[\varphi_{ed}^*, \varphi_{ex}^*]$ will supply only the international market;
3. The EU and the USA ACPs against Chinese PV will reduce the number of Chinese firms entering PV industry;
4. If China subsidies the variable production costs of PV companies, it can offset the negative impact of ACPs of the EU and the USA;
5. Increasing the upper limit of productivity in PV industry φ_{max} through technological innovation can increase the number of PV firms in China.

Appendix B. Robustness test

B.1. Placebo test

This paper employs the DID model to identify the policy impact brought by the EU and the USA ACPs. In addition to meeting the parallel trends assumption, it is also necessary to examine whether the significant differences between the treatment and control groups originate from other non-target policies or interference. Therefore, this paper conducts a placebo test by randomly drawing the treatment group and fictitious policy times. The specific method is as follows: this study randomly selects from the original full sample several countries or regions equal to the number of the original treatment group to construct a pseudo-treatment group and a pseudo-control group. At the same time, taking the importing countries or regions as the basis for grouping, a year is randomly selected within each country to obtain a fictitious policy shock time. Then, the interaction term between the pseudo-treatment group and the fictitious policy shock time is constructed for regression. This process is repeated 500 times.

The placebo test results are shown in Fig. B.7. The horizontal axis represents the regression coefficient values of the pseudo-policy variable, and the vertical axis represents the p-values and the kernel density values of the regression coefficients. The red horizontal dashed line represents the level of significance (p = 0.1), and the red vertical dashed line indicates the regression coefficient of the core independent variable in the original DID model (−2.080). It can be seen that the regression coefficients of the pseudo-policy shock are concentrated in the area near 0, and most of coefficients are above significance line (0.1) and are far from the regression coefficient value of benchmark model. This result indicates that the significant difference in PV products trade volume between the treatment group and the control group caused by the EU and the USA ACPs is not due to the influence of other random factors, and the results of benchmark regression are robust.

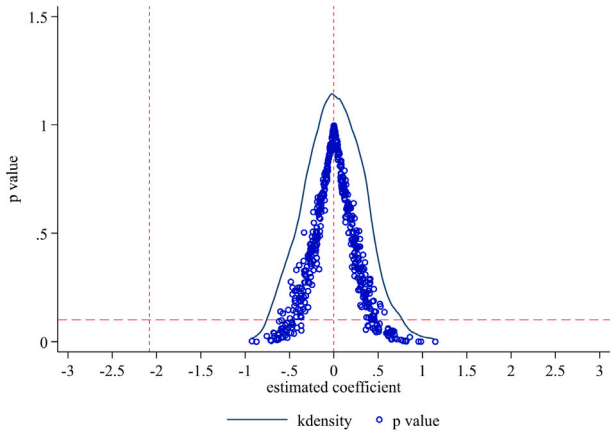


Fig. B.7. Dynamic effects analysis of new PV entrants.

B.2. Replace dependent variable and eliminate outliers

This paper evaluates the changes in Chinese PV product exports before and after ACPs of the EU and the USA by using the indicator of export value in the baseline model. Considering that the trade volume measured in currency units includes both product price and trade quantity information, this paper further replaces the dependent variable with the physical quantity of PV cells and modules exported by China in each year. Estimations are re-performed for Eqs. (1) and (2), with the results shown in Table B.6, columns (1) and (2). The study finds that, after excluding the price factor, ACPs of the EU and the USA still significantly leads to a substantial reduction in Chinese exports of PV products to EU countries and the United States. This negative impact has been present since the initiation of the investigation and has been further intensified after the affirmative final ruling. The regression coefficients are similar to the baseline regression results and are significant at 1% level.

Furthermore, considering the potential bias in the estimation of regression coefficients due to extreme values in trade statistics data, this paper applies a 1% tail trimming to the total export value of PV cells and modules from China to various countries. Regressions are then conducted based on Eqs. (1) and (2), with the results presented in Table B.6, columns (3) and (4). The findings indicate that after the exclusion of extreme values, the regression coefficients of the policy variables remain significant at 1% level. The absolute values of the regression coefficients is somewhat reduced compared to the benchmark model results, yet still confirms a significant inhibitory effect of the EU and the USA ACPs on Chinese PV product exports, demonstrating the robustness of the results.

B.3. Replacement of control group

In the benchmark regression, this paper takes Chinese PV products export as the starting point and, by comparing changes of PV cells

Table B.6
Further robustness tests.

Variable	(1)	(2)	(3)	(4)	(5)	(6)
$Treat_i \times Poltime_{it}^1$	−2.239***		−2.011***		−1.508***	
	(0.346)		(0.305)		(0.047)	
$Treat_i \times Poltime_{it}^2$		−1.175***		−1.137***		−0.484***
		(0.353)		(0.318)		(0.034)
$Treat_i \times Poltime_{it}^3$		−2.844***		−2.452***		−1.850***
		(0.371)		(0.337)		(0.054)
Control	YES	YES	YES	YES	YES	YES
Year	YES	YES	YES	YES	YES	YES
Country/region	YES	YES	YES	YES	YES	YES
Constant	−6.004	4.990	0.029	6.934	125.917***	126.650***
	(17.040)	(17.186)	(12.466)	(12.757)	(20.547)	(20.612)
adj. R2	0.815	0.821	0.824	0.828	0.938	0.939
N	1704	1704	1858	1858	2889	2889

and modules trade between China and Europe and the USA, and that between China and other economies, before and after the policy implementation, the study confirms the trade-suppressing effect caused by ACPs of the EU and the USA. To clearly and comprehensively delineate the policy effects, this paper attempts to adjust the research perspective, focusing on the trade of products between China -EU and China- USA, using product categories based on HS code as the basis for classification of treatment and control groups. The export of PV cells and modules from China to Europe and to the USA is considered the treatment group, while other electrical machinery and equipment exported from China to these two regions serve as the control group. The paper still examines how the policy effect of ACPs of the EU and the USA is manifested.

Referring to "Import and Export Tariff of the People's Republic of China (2019)" issued by the Tariff Commission of the State Council, it is observed that PV cells and modules are classified under the HS 85 category, which, along with HS 84, is aggregated under the sixteenth section of the tariff, pertaining to machinery, electrical equipment, and their respective components. Given the similar nature of these commodities, the study introduces a novel dataset that excludes the HS 8541 category, traditionally associated with PV cells and modules, to establish a control group. Concurrently, the export of these specific commodities to the European Union and the United States serves as the treatment group. The methodology incorporates standard error clustering by product category, while maintaining consistency with the benchmark model for other variables.

Table B.6 columns (5) and (6) presents the impact of the EU and the USA ACPs on PV products export from China to Europe and the United States, as well as on the export of other electrical machinery products, where the treatment group has been replaced. Results indicate that the implementation of ACPs has indeed led to a significant reduction in Chinese PV cells and modules export to these regions. The regression coefficients remain statistically significant at 1% level.

B.4. Replace regression model

The core dependent variable in this paper is trade volume, which may lead to biased results when using OLS regression model due to the presence of zeros in such data. In light of this, the paper further employs a Poisson regression model and incorporates all variables from the baseline regression for robustness test. The results are presented in Table B.7 column(1)–(3). When accounting for the zero-value existence, all policy variables remain significantly negative at 1% level, with coefficient values showing no significant difference compared to the baseline regression, indicating that the robustness of baseline results.

B.5. Refinement of control variables

In this study, some control variables within the regression model exhibit missing values, potentially skewing the results due to data incompleteness. To address this issue, this paper performs interpolation on the control variables with missing values, and incorporated the new control variables in the model for subsequent regression analysis. The findings reveal that, after accounting for the impact of missing data in certain control variables, the negative effect of ACPs on China's photovoltaic product exports remains significantly pronounced (as shown in Table B.7 column(4)–(5)), indicating that the baseline regression results are robust.

B.6. Differential impact of ACPs measures for the US and for the EU

To test whether the ACPs of different economies have different policy effects, this paper has further distinguished the ACPs of the EU and the USA, as well as the samples, conducting separate regression analyses for EU versus non-EU/USA samples and US versus non-EU/USA samples. The results (as shown in Table B.8) reveal that after distinguishing between the EU and the US, the ACPs of both economies have individually exerted a significant suppressive effect on China's PV exports. Notably, when the model incorporates both investigations and final rulings period, the policy effects of the EU's two time points are significantly negative, while the USA only shows a significant negative policy effect at final ruling time point. Additionally, the negative effect of the EU ACPs is more pronounced.

B.7. Identification of PV-related enterprises

We identify photovoltaic-related companies through keyword searches, which is reasonable in principle but could still lead to false positives or omissions. To verify the robustness of our findings, we employ more precise identification criteria for PV enterprises. First, from our baseline sample of PV-related firms, we exclude companies classified under "retail" and "wholesale" sectors. The new regression results are presented in columns (1) and (2) of Table B.9. Second, building upon this refined sample, we further exclude firms whose business scope includes "solar water heaters" "storage batteries" and "mobile phone batteries". The corresponding regression results are shown in columns (3) and (4) of Table B.9. The results indicate that even after excluding these potentially unrelated enterprises, the implementation of ACPs of the EU and the USA still shows no negative impact on the number of new PV enterprises in China.

Appendix C. Discussion from the perspective of PV-related patent application

This paper's theoretical analysis indicates that the ability to persist in innovative research and development, and maintain technological advantages is an important factor affecting industry recovery from external shocks. Therefore, based on national patent application data from 2009 to 2018, this paper uses PV-related patent applications in China as a quantified indicator of innovative R&D intensity in Chinese PV industry. It further examines whether the innovative activities in

Table B.7
Replace regression model and Refinement of control variables.

Variable	(1) val_pv	(2) val_pv	(3) val_pv	(4) Logval_pv	(5) Logval_pv
$Treat_i \times Poltime_{it}^1$	−2.261***	−1.114***		−2.170***	
	(0.563)	(0.396)		(0.391)	
$Treat_i \times Poltime_{it}^2$			−0.655**		−0.894**
			(0.327)		(0.385)
$Treat_i \times Poltime_{it}^3$			−1.641***		−2.674***
			(0.463)		(0.427)
Control	NO	YES	YES	YES	YES
Year	YES	YES	YES	YES	YES
Country/region	YES	YES	YES	YES	YES
Constant	14.075***	−34.448	−26.262	7.450***	7.508***
	(0.125)	(38.999)	(30.670)	(0.122)	(0.115)
pseudo R2/adj.R2	0.881	0.934	0.940	0.791	0.797
N	2276	1997	1997	2001	2001

Table B.8
Differential impact of ACPs measures for the US and for the EU.

Variable	EU			USA		
	(1)	(2)	(3)	(4)	(5)	(6)
	Logval_pv					
$Treat_i \times Poltime_{it}^1$	−2.363***	−2.025***		−1.267***	−1.602***	
	(0.407)	(0.315)		(0.118)	(0.120)	
$Treat_i \times Poltime_{it}^2$			−1.215***			−0.020
			(0.326)			(0.256)
$Treat_i \times Poltime_{it}^3$			−2.458***			−2.139***
			(0.353)			(0.156)
Control	No	YES	YES	No	YES	YES
Year	YES	YES	YES	YES	YES	YES
Country/region	YES	YES	YES	YES	YES	YES
Constant	7.559***	1.359	7.622	7.045***	5.311	5.679
	(0.045)	(12.361)	(12.708)	(0.001)	(13.044)	(13.032)
adj.R2	0.786	0.823	0.826	0.799	0.823	0.823
N	2014	1848	1848	1716	1556	1556

Table B.9
Identification of PV-Related enterprises.

Variable	(1) entnum	(2) entnum	(3) entnum	(4) entnum
Did	0.279***	0.262***	0.425***	0.408***
	(0.030)	(0.027)	(0.032)	(0.030)
Control	NO	YES	NO	YES
Year	YES	YES	YES	YES
County	YES	YES	YES	YES
Constant	5.939***	5.883***	5.939***	5.883***
	(0.000)	(0.088)	(0.000)	(0.088)
pseudo R2	0.933	0.933	0.933	0.933
N	832 016	832 016	832 008	832 008

Chinese PV industry were negatively affected after the introduction and implementation of ACPs of the EU and the USA. The aim is to reveal why Chinese PV industry can emerge from the shadow of ACPs. Similarly, when using the number of PV-related and non-PV-related patents in each county as the dependent variable, this paper employs Poisson regression to estimate Eq. (4). The regression results are shown in Table C.10, where columns (1) to (3) categorize (non-)PV-related patents based on the inclusion of one of 18 keywords in patent abstract, and columns (4) to (6) identify (non-)PV-related patents based on patent classification codes. The results of each column indicate that after the execution of ACPs of the EU and the USA, the number of PV-related patents in Chinese counties increased significantly by more than 10% compared to non-PV-related patents. This suggests that the ACPs did not hinder the innovative R&D of Chinese PV industry.

It should be noted that the regression results in Table C.10 only indicate that ACPs of the EU and the USA did not have a negative impact on the innovative activities of Chinese PV industry; while they do not imply that ACPs promoted innovation in the Chinese PV industry. In fact, in the same year that Europe and the United States initiated ACPs investigations, Chinese PV patent applications surpassed those of the USA for the first time, ranking first in the world.[18] Since then, over the following decade, Chinese PV patent applications have continued to grow rapidly. As of May 2022, ten years after 2012, Chinese PV patent applications accounted for 80.14% of global total PV patent applications,[19] significantly leading other countries. Europe and the United States chose to implement ACPs instead of selecting a technological innovation path to enhance the competitiveness of their own PV industries. In contrast, China insisted on breaking through trade barriers with technological innovation. This starkly contrasting choice of paths ultimately prompted the Chinese PV industry to successfully respond to severe external challenges, demonstrating strong resilience in industrial development. The past decade's rebirth of Chinese PV industry development has become a classic case of technological innovation overcoming trade barriers.

[18] http://www.cima.org.cn/nnews.asp?vid=36003.
[19] https://solar.ofweek.com/2022-06/ART-260006-8420-30564561.html.

Table C.10
Changes in PV-related patent applications against the background of ACPs of the EU and the USA.

Variable	(1) patent1	(2) patent1	(3) patent1	(4) patent2	(5) patent2	(6) patent2
Did	0.141***	0.144***	0.118***	0.159***	0.176***	0.172***
	(0.024)	(0.025)	(0.021)	(0.024)	(0.023)	(0.023)
Control	NO	YES	YES	NO	YES	YES
Year	YES	YES	YES	YES	YES	YES
County	YES	YES	YES	YES	YES	YES
County-year	NO	NO	YES	NO	NO	YES
Constant	8.032***	8.159***	393.742***	7.891***	8.064***	382.342***
	(0.001)	(0.140)	(145.962)	(0.004)	(0.133)	(138.597)
pseudo R2	0.972	0.973	0.984	0.969	0.970	0.982
N	50 728	50 351	50 351	50 991	50 611	50 611

Data availability

Data will be made available on request.

References

Beck, T., Levine, R., Levkov, A., 2010. Big bad banks? the winners and losers from bank deregulation in the United States. J. Financ. 65 (5), 1637–1667.

Biasi, B., Sarsons, H., 2021. Flexible wages, bargaining, and the gender gap*. Q. J. Econ. 137 (1), 215–266.

Binz, C., Tang, T., Huenteler, J., 2017. Spatial lifecycles of cleantech industries – The global development history of solar photovoltaics. Energy Policy 101, 386–402.

Blonigen, B., Prusa, T., 2016. Dumping and antidumping duties. In: Bagwell, K., Staiger, R.W. (Eds.), Handbook of Commercial Policy. Vol. 1, North-Holland, pp. 107–159.

Borusyak, K., Jaravel, X., Spiess, J., 2024. Revisiting event-study designs: Robust and efficient estimation. Rev. Econ. Stud. rdae007.

Chen, Z., Yu, B., Yang, C., Zhou, Y., Yao, S., Qian, X., Wang, C., Wu, B., Wu, J., 2021. An extended time series (2000–2018) of global NPP-VIIRS-like nighttime light data from a cross-sensor calibration. Earth Syst. Sci. Data 13 (3), 889–906.

Chen, Y., Zhou, J., Ge, Y., Dong, J., 2024. Uncovering the rapid expansion of photovoltaic power plants in China from 2010 to 2022 using satellite data and deep learning. Remote Sens. Environ. 305, 114100.

Cheng, L., Mi, Z., Coffman, D., Meng, J., Chang, D., 2021. Destruction and deflection: Evidence from American antidumping actions against China. Struct. Change Econ. Dyn. 57, 203–213.

Cohn, J.B., Liu, Z., Wardlaw, M.I., 2022. Count (and count-like) data in finance. J. Financ. Econ. 146 (2), 529–551.

Crowley, M., Meng, N., Song, H., 2018. Tariff scares: Trade policy uncertainty and foreign market entry by Chinese firms. J. Int. Econ. 114, 96–115.

Cui, L., Wang, J., Chen, X., Song, M., 2024. Regional policy options for carbon peaking in the Yangtze River Delta under uncertainty. J. Environ. Manag. 364, 121445.

Felbermayr, G., Sandkamp, A., 2020. The trade effects of anti-dumping duties: Firm-level evidence from China. Eur. Econ. Rev. 122, 103367.

Fontagné, L., Orefice, G., Piermartini, R., Rocha, N., 2015. Product standards and margins of trade: Firm-level evidence. J. Int. Econ. 97 (1), 29–44.

Gardner, J., 2022. Two-stage differences in differences. arXiv:2207.05943.

Goodman-Bacon, A., 2021. Difference-in-differences with variation in treatment timing. J. Econometrics 225 (2), 254–277, Themed Issue: Treatment Effect 1.

Grundke, R., Moser, C., 2019. Hidden protectionism? Evidence from non-tariff barriers to trade in the United States. J. Int. Econ. 117, 143–157.

Huang, Y., Lin, C., Liu, S., Tang, H., 2023. Trade networks and firm value: Evidence from the U.S.-China trade war. J. Int. Econ. 145, 103811.

Jabbour, L., Tao, Z., Vanino, E., Zhang, Y., 2019. The good, the bad and the ugly: Chinese imports, European union anti-dumping measures and firm performance. J. Int. Econ. 117, 1–20.

Kao, K.-F., Peng, C.-H., 2016. Anti-dumping protection, price undertaking and product innovation. Int. Rev. Econ. Financ. 41, 53–64.

Kong, D., Liu, C., Narayan, P.K., Sharma, S.S., 2024. The US–China trade war and corporate innovation: Evidence from China. Financ. Manag. 53 (3), 501–541.

Krugman, P.R., 1979. Increasing returns, monopolistic competition, and international trade. J. Int. Econ. 9 (4), 469–479.

Krugman, P., 1980. Scale economies, product differentiation, and the pattern of trade. Am. Econ. Rev. 70 (5), 950–959.

Li, J., Lu, Y., Song, H., Xie, H., 2019. Long-term impact of trade liberalization on human capital formation. J. Comp. Econ. 47 (4), 946–961.

Li, Q., Qian, T., Wang, J., Long, R., Chen, H., Sun, C., 2023. Social "win-win" promotion of green housing under the four-subject evolutionary game. Energy Econ. 127, 107117.

Li, C., Whalley, J., 2021. Trade protectionism and US manufacturing employment. Econ. Model. 96, 353–361.

Liu, X., Cifuentes-Faura, J., Zhao, S., Wang, L., 2023. Government environmental attention and carbon emissions governance: Firm-level evidence from China. Econ. Anal. Policy 80, 121–142.

Liu, M., Shan, Y., Li, Y., 2022. Study on the effect of carbon trading regulation on green innovation and heterogeneity analysis from China. Energy Policy 171, 113290.

Lu, Y., Tao, Z., Zhang, Y., 2013. How do exporters respond to antidumping investigations? J. Int. Econ. 91 (2), 290–300.

McCarthy, K.J., 2016. On the influence of the European trade barrier on the Chinese pv industry: Is the solution to the solar-dispute "successful"? Energy Policy 99, 154–157.

Melitz, M.J., 2003. The impact of trade on intra-industry reallocations and aggregate industry productivity. Econom. 71 (6), 1695–1725.

Metiu, N., 2021. Anticipation effects of protectionist U.S. Trade policies. J. Int. Econ. 133, 103536.

Miller, D.L., 2023. An introductory guide to event study models. J. Econ. Perspect. 37 (2), 203–230.

Nelson, K.P., Parton, L.C., Brown, Z.S., 2022. Biofuels policy and innovation impacts: Evidence from biofuels and agricultural patent indicators. Energy Policy 162, 112767.

Nguyen, L., Kinnucan, H.W., 2019. The US solar panel anti-dumping duties versus uniform tariff. Energy Policy 127, 523–532.

Rambachan, A., Roth, J., 2023. A more credible approach to parallel trends. Rev. Econ. Stud. 90 (5), 2555–2591.

Roth, J., 2022. Pretest with caution: Event-study estimates after testing for parallel trends. Am. Econ. Rev.: Insights 4 (3), 305–322.

Signoret, J., Cieszkowsky, M., Erbahar, A., 2020. Temporary trade barriers database including the global antidumping database.

Sun, L., Abraham, S., 2021. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. J. Econometrics 225 (2), 175–199, Themed Issue: Treatment Effect 1.

Sun, C., Khan, A., Cai, W., 2024. The response of energy aid and natural resources consumption in load capacity factor of the Asia Pacific emerging countries. Energy Policy 190, 114150.

Sun, C., Min, J., 2024. Dynamic trends and regional differences of economic effects of ultra-high-voltage transmission projects. Energy Econ. 138, 107871.

Vandenbussche, H., Viegelahn, C., 2018. Input reallocation within multi-product firms. J. Int. Econ. 114, 63–79.

Wang, X., Liu, Z., Lv, Y., Zhao, C., 2019. Trade barriers and participation in the global value chain: An empirical study based on Anti-dumping toward China. China World Econ. 27 (2), 86–106.

Wang, X., Shi, B., 2022. Transmission and diffusion effect of Sino-US trade friction along global value chains. Financ. Res. Lett. 46, 102057.

Wang, S., Shu, W., Cui, L., 2024. Green finance policy and green economic transformation. Int. Rev. Financ. Anal. 96, 103785.

Xia, Y., Kong, Y., Ji, Q., Zhang, D., 2019. Impacts of China-US trade conflicts on the energy sector. China Econ. Rev. 58, 101360.